Math 196 Lecture Notes: Week 9

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0.1 Eigenvectors and Eigenvalues

Let A be an $n \times n$ matrix. A *eigenvector* of A is a vector v such that $Av = \lambda v$ for some scalar λ, the *eigenvalue*. If λ is the eigenvalue of a nonzero eigenvector of A, we say that it is an eigenvalue of A.

0.2 The Characteristic Polynomial

We can say that λ is an eigenvalue of A if and only if $A - \lambda I_n$ has a nonzero kernel. This happens exactly when $\det(A - \lambda I_n) = 0$, or equivalently, when $p_A(\lambda) = 0$, for $p_A(t) = \det(A - tI_n)$. We call p_A the *characteristic polynomial* of A. Its roots are exactly the eigenvalues.

0.3 Eigenspaces

For an eigenvalue λ, the set of all eigenvectors with eigenvalue λ forms a linear subspace $V_\lambda = \ker(A - \lambda I_n)$. We call this an *eigenspace*. Thus there are infinitely many eigenvectors for each eigenvalue.

0.4 Diagonalization

If $v_1, \ldots, v_n$ is a basis consisting of eigenvectors for A, so $Av_i = \lambda_i v_i$, and

$$S = \begin{bmatrix} \vdots & & \vdots \\ v_1 & \ldots & v_n \\ \vdots & & \vdots \end{bmatrix}$$

Then we can write A as

$$A = S \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} S^{-1}$$

This is called a *diagonalization* of A.

Question. Given a diagonalization of A, how can we tell if A is invertible? If it is, what is the inverse?

Example 1. Consider the reflection matrix

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

This has eigenvectors

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

with eigenvalues $\lambda_1 = 1$, $\lambda_2 = -1$. The eigenvectors form a basis for $\mathbb{R}^2$, and so we have the following diagonalization of A:

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}^{-1}$$

Example 2. Consider the matrix

$$\begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$$

This has characteristic polynomial

$$\det \begin{bmatrix} 1-t & 2 \\ 4 & 3-t \end{bmatrix} = t^2 - 4t + 3 - 8 = t^2 - 4t - 5 = (t-5)(t+1)$$

and so the eigenvalues are 5 and -1. The eigenspaces are

$$V_5 = \ker \begin{bmatrix} -4 & 2 \\ 4 & -2 \end{bmatrix} = \operatorname{span}\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$$

$$V_{-1} = \ker \begin{bmatrix} 2 & 2 \\ 4 & 4 \end{bmatrix} = \operatorname{span}\left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$$

Thus we have a diagonalization

$$\begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 5 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix}^{-1}$$

1 Criteria for Diagonalizability

1.1 When is a matrix diagonalizable?

We say that a matrix is diagonalizable if it has a diagonalization. From the above remarks, we can say that a matrix is diagonalizable if and only if there is a basis of $\mathbb{R}^n$ consisting of eigenvectors for A. The following proposition makes it more clear what needs to be true for such a basis to exist.

Theorem 1. *Suppose that we choose bases for distinct eigenspaces $V_{\lambda_1}, \dots, V_{\lambda_k}$, and put them together to get a list of vectors $v_1, \dots, v_r$. This list of vectors is linearly independent.*

I will prove a simpler version of this; on the homework, I will ask you to deduce the above statement from this special case.

Proposition. *Suppose $v_1, \dots, v_r$ are vectors such that $v_i \in V_{\lambda_i}$ for distinct eigenvalues $\lambda_1, \dots, \lambda_r$. Then if $v_1 + \cdots + v_r = 0$, it must be that each $v_i = 0$.*

Proof. Suppose that $v_1 + \cdots + v_r = 0$, and some v_i is not 0. We can omit the ones which are 0, and so assume that every $v_i \neq 0$. Applying the matrix $A - \lambda_r I_n$ to both sides, we get

$$(\lambda_1 - \lambda_r)v_1 + \cdots + (\lambda_{r-1} - \lambda_r)v_{r-1} = 0$$

Each constant $\lambda_i - \lambda_r$ for $i < r$ is nonzero, so we have reduced the number of vectors in the sum, keeping the remaining ones nonzero. We can keep doing this until we get $k \cdot v_1 = 0$ for some nonzero k, and so $v_1 = 0$, a contradiction to what we assumed. □

Corollary (of the theorem)**.** *An $n \times n$ matrix A with eigenvalues $\lambda_1, \dots, \lambda_k$ is diagonalizable if and only if*

$$\dim V_{\lambda_1} + \cdots + \dim V_{\lambda_k} = n$$

This condition means that putting together bases for each eigenspace, we get a basis for $\mathbb{R}^n$.

By definition, for any eigenvalue λ, $\dim V_\lambda \geq 1$. A consequence of this is that if a matrix A has n distinct eigenvalues, it is diagonalizable. We can get an eigenbasis by choosing one nonzero eigenvector for each eigenvalue.

Example 3. We saw previously that the matrix

$$R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

has no (real) eigenvalues, and so it is not diagonalizable. (Later, we will see that this matrix becomes diagonalizable if we work with complex numbers.)

Example 4. Consider the matrix

$$G = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

This has characteristic polynomial

$$p_G(t) = \det \begin{bmatrix} 1-t & 1 \\ 0 & 1-t \end{bmatrix} = (1-t)^2$$

Its only eigenvalue is 1. We can check that

$$V_1 = \ker \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \operatorname{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$$

has dimension 1. Therefore, G is not diagonalizable either. (This one will not be diagonalizable even if we use complex numbers.)

Example 5. Consider the matrix

$$H = \begin{bmatrix} 1 & 2 & 1 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

from before. Recall that for this matrix,

$$V_1 = \operatorname{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

$$V_{-2} = \operatorname{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} \right\}$$

From the theorem above, if we put these vectors together, we get a basis

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix}$$

which diagonalizes H:

$$H = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -3 & -2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -3 & -2 \end{bmatrix}^{-1}$$

1.2 Multiplicities of Eigenvalues

In the previous 2 examples, the characteristic polynomial had repeated roots. Let's introduce some terminology to deal with this.

Definition. For a polynomial $f(t)$ with a root λ, the *multiplicity* of λ as a root of f is the largest m such that $(t-\lambda)^m$ appears as a factor of $f(t)$. If λ is an eigenvalue of a matrix A, its *multiplicity* is its multiplicity as a root of $p_A(t)$.

We will see that this is related to the dimension of the corresponding eigenspaces. However, we first need to understand how eigenvalues of similar matrices are related.

Proposition. *Suppose that A and B are similar. Then $p_A(t) = p_B(t)$.*

Proof. Let $B = S^{-1}AS$. Then

$$B - tI_n = S^{-1}AS - tI_n = S^{-1}AS - S^{-1}tI_nS = S^{-1}(A - tI_n)S$$

This is to say $B - tI_n$ and $A - tI_n$ are similar. Because similar matrices have the same determinant, $p_B(t) = p_A(t)$. ☐

Corollary. *Similar matrices have the same eigenvalues with the same multiplicities.*

Theorem 2. *If λ is an eigenvalue of A of multiplicity m, then*

$$\dim(V_\lambda) \leq m$$

Proof. Let $v_1, \ldots, v_r$ be a basis for V_λ. This can be extended to a basis $v_1, \ldots, v_n$ for $\mathbb{R}^n$. Letting

$$S = \begin{bmatrix} \vdots & & \vdots \\ v_1 & \ldots & v_n \\ \vdots & & \vdots \end{bmatrix}$$

we have

$$S^{-1}AS = \begin{bmatrix} \lambda & & & * \\ & \ddots & & * \\ & & \lambda & * \\ 0 & \ldots & 0 & C \end{bmatrix} = B$$

One can check that $p_A(t) = p_B(t) = (\lambda - t)^r p_C(t)$ (think about taking a cofactor expansion of $\det(B - tI_n)$), and so $m \geq r$. ☐

We can summarize the criteria for diagonalizability as follows:

1. Given an $n \times n$ matrix A, it has a characteristic polynomial $p_A(t)$. Its eigenvalues $\lambda_1, \ldots, \lambda_k$ are the roots of this. Counted with multiplicities $m_1, \ldots, m_k$, there are at most n of them; $m_1 + \cdots + m_k \leq n$.
2. In order for the matrix to be diagonalizable, we must have $\dim V_{\lambda_1} + \cdots + \dim V_{\lambda_k} = n$. In order for this to be true, two things must occur. First, the characteristic polynomial must have enough roots, in the sense that $m_1 + \cdots + m_k = n$.
3. Second, for each λ_j, we must have $\dim V_{\lambda_j} = m_j$.
4. If these conditions are met, then we can combine bases of all the V_{λ_j} to get a basis for $\mathbb{R}^n$, this yields a diagonalization $A = SDS^{-1}$ of the original matrix A.

1.3 Complex Eigenvalues

As we have seen, some matrices, like

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

fail to be diagonalizable because the characteristic polynomial, in this case $t^2 + 1$ fails to have enough real roots. On the other hand, this polynomial has complex roots $t = \pm i$. More generally, if we look for roots in $\mathbb{C}$, we always have enough:

Theorem 3 (Fundamental Theorem of Algebra)**.** *A degree n polynomial always has n roots in $\mathbb{C}$, counted with multiplicity.*

To realize these roots as eigenvalues, we need to consider vectors with complex entries as well. We denote the set of such vectors with n entries $\mathbb{C}^n$. Almost all of the results we have for vectors in $\mathbb{R}^n$, linear transformations, matrices, and determinants also apply in $\mathbb{C}^n$, with multiplication by complex scalars replacing multiplication by real scalars. In particular, the notions of subspaces, linear dependence, linear transformations, and dimension work the same way. All the properties of $\mathbb{R}$ that we used, that we can add, multiply, subtract, and divide by nonzero numbers, are also true for $\mathbb{C}$. To clarify when we are working with the complex version of things like dimension, I may write $\dim_{\mathbb{C}}(V)$, for $V \subset \mathbb{C}^n$ a complex subspace, in contrast to $\dim_{\mathbb{R}}(W)$ for $W \subset \mathbb{R}^n$ a real subspace. In particular $\dim_{\mathbb{C}}(\mathbb{C}^1) = 1$, even if you are used to thinking of $\mathbb{C}$ as a 2 dimensional object.

Remark. The following formula is useful for carrying out computations with complex numbers:

$$\frac{1}{a+bi} = \frac{a-bi}{(a+bi)(a-bi)} = \frac{a-bi}{a^2+b^2}$$

There is one exception to the statement that everything works the same way over $\mathbb{C}$, which is the dot product. To see what changes, if we define $v \cdot w$ by the same formula, we can see that

$$\begin{bmatrix} 1 \\ i \end{bmatrix} \cdot \begin{bmatrix} 1 \\ i \end{bmatrix} = 1^2 + i^2 = 0$$

and so we would be inclined to say that this vector has 0 length, even though it is nonzero. There is a way to modify the definition of dot product so that it can be used in the complex world, but for this course we will not need this. In general, we will not mention anything related to length, angle, or orthogonality when we are thinking about vectors in $\mathbb{C}^n$.

With this said, we can apply our previous methods of finding eigenvalues and eigenvectors in the complex world, with the benefit that now our characteristic polynomial will have as many zeroes as we could hope for.

Example 6. Consider again the rotation matrix

$$R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

with characteristic polynomial $p_r(t) = t^2 + 1$. The eigenvalues are the zeroes of this polynomial, $\pm i$. Let's find the corresponding eigenvectors. In other words, we want to find vectors in the kernel of the following matrices:

$$\begin{bmatrix} -i & -1 \\ 1 & -i \end{bmatrix}$$

To find such a vector, we bring the matrix to reduced row echelon form as before:

$$\begin{bmatrix} -i & -1 \\ 1 & -i \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & -i \\ -i & -1 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & -i \\ 0 & 0 \end{bmatrix}$$

This gives us an equation $x = iy$, and so a general element of the kernel has the form

$$\begin{bmatrix} iy \\ y \end{bmatrix} = y \begin{bmatrix} i \\ 1 \end{bmatrix}$$

This is our eigenvector for i. For $-i$, we use the same method:

$$\begin{bmatrix} i & -1 \\ 1 & i \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & i \\ i & -1 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & i \\ 0 & 0 \end{bmatrix}$$

This gives the equation $x = -iy$, and so a general element of the kernel is a scalar multiple of

$$\begin{bmatrix} -i \\ 1 \end{bmatrix}$$

This is our eigenvector for $-i$. Thus we have the following diagonalization of R:

$$R = \begin{bmatrix} i & -i \\ 1 & 1 \end{bmatrix} \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} \begin{bmatrix} i & -i \\ 1 & 1 \end{bmatrix}^{-1}$$

Allowing for the use of complex numbers, our criteria for diagonalization become simpler:

1. Working in $\mathbb{C}$, the characteristic polynomial $p_A(t)$ always factors completely as

$$(\lambda_1 - t)^{m_1} \dots (\lambda_k - t)^{m_k}$$

   where $\lambda_i \in \mathbb{C}$ are the eigenvalues, and $m_1 + \cdots + m_k = n$.

2. For each j, the eigenspace $V_{\lambda_j} \subset \mathbb{C}^n$ has dimension $\dim_{\mathbb{C}} V_{\lambda_j} \leq m_j$.

3. If for every eigenvalue, we have $\dim_{\mathbb{C}} V_{\lambda_j} = m_j$, then the matrix is diagonalizable. Otherwise, it is not diagonalizable. In the case that it is diagonalizable, we can find a diagonalization by putting together bases for each V_{λ_j}.

1.4 Orthogonal Matrices in Dimension Three

Before moving on to applications of diagonalization per se, I would like to discuss another application of eigenvectors and eigenvalues, which is to classify all orthogonal transformations of $\mathbb{R}^3$.

Let's first see what orthogonal transformations look like in $\mathbb{R}^2$. For a 2×2 orthogonal matrix A, the first column must have the form

$$\begin{bmatrix} a \\ b \end{bmatrix}$$

where $a^2 + b^2 = 1$. The second column is a unit vector orthogonal to the first. There are only two such vectors [draw picture],

$$\begin{bmatrix} -b \\ a \end{bmatrix}, \begin{bmatrix} b \\ -a \end{bmatrix}$$

These correspond to the cases that $\det(A) = 1$, $\det(A) = -1$ respectively. In the first case, taking $a = \cos(\theta)$, $b = \sin(\theta)$, we have

$$A = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

is a rotation matrix. For the second case

$$A = \begin{bmatrix} a & b \\ b & -a \end{bmatrix}$$

the characteristic polynomial is $t^2 - 1$, and so we have eigenvalues $1, -1$. The 1-eigenspace is spanned by

$$\begin{bmatrix} -b \\ a - 1 \end{bmatrix}$$

and one can check that A is a reflection across this line.

Now we consider dimension 3. Recall that an orthogonal matrix can have determinant $+1$ or -1. Any 3×3 matrix has at least one real eigenvalue, and because orthogonal matrices are length preserving, the real eigenvalues must be ± 1. We can relate the eigenvalues to the determinant:

Proposition. *For a matrix A with complex eigenvalues $\lambda_1, \dots, \lambda_r$ with multiplicities $m_1, \dots, m_r$, we have* $\det(A) = \lambda_1^{m_1} \dots \lambda_r^{m_r}$.

Proof. Plug in $t = 0$ in the equation

$$\det(A - tI_n) = (\lambda_1 - t)^{m_1} \dots (\lambda_r - t)^{m_r}$$

□

The final thing we need is that complex (i.e not real) eigenvalues, being complex roots of a polynomial with real coefficients, come in conjugate pairs $z, \overline{z}$, and $z\overline{z} = |z|^2 > 0$. Now suppose we have an orthogonal 3×3 matrix A with $\det(A) = 1$. I will show that A is a rotation about some axis. Using the fact that A has at least one real eigenvalue and the product of the eigenvalues is 1, we have the following possibilities:

$$1, z, \overline{z}$$

$$1, -1, -1$$

$$1, 1, 1$$

In all cases we have an eigenvalue 1. An eigenvector u for this gives us a line fixed by A, this will be the axis of rotation. [Draw picture] Because A is orthogonal, it takes the plane orthogonal to this line to itself. [add to picture]. We can choose an orthonormal basis u, v, w where v and w span this plane. Changing coordinates to this basis, the transformation now looks like

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & a & b \\ 0 & c & d \end{bmatrix}$$

where $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is orthogonal with determinant 1. This must then be a rotation matrix, and so A is a rotation about the line spanned by u. In the case that $\det(A) = -1$, a similar analysis can be used to show that A represents a rotation as above, followed by a reflection across the plane.

To follow up on something I said at the very beginning of the course, if we have two 3×3 rotation matrices A and B, then their product AB is orthogonal and has determinant 1, so is a rotation. This is to say that one rotation followed by another yields a rotation about some third axis. We can find the axis of rotation by finding the eigenspace V_1 of AB.

How do we find the angle of rotation? Well given any 3×3 orthogonal matrix A, we know that A is similar to a matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) \\ 0 & \sin(\theta) & \cos(\theta) \end{bmatrix}$$

[refer to previous picture for the coordinate system]. Thus the eigenvalues of A are 1 together with the eigenvalues of the rotation

$$\begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

which you can check are the (in general) complex numbers $\cos(\theta) \pm i\sin(\theta)$. This does not allow us to distinguish between θ and $-\theta$, but this is to be expected as the direction of rotation depends on a choice of direction around the axis.

In summary, given any 3×3 orthogonal matrix A, we know that A is a rotation about some axis. To find the axis of rotation, find the eigenspace V_1. To find the angle of rotation (at least up to sign) find the remaining (probably complex) eigenvalues, and compute $\cos^{-1}$ of their real part. We can in particular apply this to compositions of rotations about different axes.

2 Linear Recurrences

Our first application of diagonalization will be to linear recurrences. A linear recurrence is a rule describing a sequence of numbers, expressing the nth term a_n as a linear function of the previous terms $a_{n-1}, a_{n-2}, \ldots$. An example you may have seen is the *Fibonacci sequence*, which is defined by the rules

$$a_0 = a_1 = 1$$

$$a_n = a_{n-1} + a_{n-2} \; : \; n > 1$$

With this rule, we can work out the terms of the sequence one by one:

$$1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, \ldots$$

Considering such a sequence, we can ask for an explicit formula for a_n in terms of n, or ask about the long term rate of growth. Diagonalization will allow us to answer both questions. First of all, we can write the linear recurrence concisely in terms of matrix multiplication. For the Fibonacci sequence, we have the rule

$$\begin{bmatrix} a_n \\ a_{n+1} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} a_{n-1} \\ a_n \end{bmatrix}$$

This gives us a formula

$$\begin{bmatrix} a_n \\ a_{n+1} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}^n \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

This formula becomes much more transparent if we diagonalize the matrix

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$

In general, if $A = SDS^{-1}$, then $A^2 = SDS^{-1}SDS^{-1} = SD^2S^{-1}$, and more generally $A^n = SD^nS^{-1}$. Computing powers of a diagonal matrix is simply a matter of computing powers of the diagonal entries. For our particular A, the characteristic polynomial is

$$\det \begin{bmatrix} -t & 1 \\ 1 & 1-t \end{bmatrix} = t^2 - t - 1$$

This has roots

$$t = \frac{1 \pm \sqrt{5}}{2}$$

We use φ to denote $\frac{1+\sqrt{5}}{2} \approx 1.6$. This number, the so called golden ratio, satisfies lots of identities, like $1 - \varphi = -\varphi^{-1}$, which make the diagonalization of this matrix look nice. The other root is $-\varphi^{-1} = \frac{1-\sqrt{5}}{2} \simeq -0.6$. For the eigenvalue φ, our eigenspace is

$$V_\varphi = \ker \begin{bmatrix} -\varphi & 1 \\ 1 & -\varphi^{-1} \end{bmatrix} = \operatorname{span} \left\{ \begin{bmatrix} 1 \\ \varphi \end{bmatrix} \right\}$$

The other eigenspace is

$$V_{-\varphi^{-1}} = \ker \begin{bmatrix} \varphi^{-1} & 1 \\ 1 & \varphi \end{bmatrix} = \operatorname{span} \left\{ \begin{bmatrix} -\varphi \\ 1 \end{bmatrix} \right\}$$

This gives us a diagonalization

$$A = \begin{bmatrix} 1 & -\varphi \\ \varphi & 1 \end{bmatrix} \begin{bmatrix} \varphi & 0 \\ 0 & -\varphi^{-1} \end{bmatrix} \begin{bmatrix} 1 & -\varphi \\ \varphi & 1 \end{bmatrix}^{-1}$$

We then have

$$\begin{bmatrix} a_n \\ a_{n+1} \end{bmatrix} = \begin{bmatrix} 1 & -\varphi \\ \varphi & 1 \end{bmatrix} \begin{bmatrix} \varphi^n & 0 \\ 0 & (-\varphi^{-1})^n \end{bmatrix} \begin{bmatrix} 1 & -\varphi \\ \varphi & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Multiplying this out, we get *Binet's formula*

$$a_n = \frac{\varphi^{n+1} - (-\varphi^{-1})^{n+1}}{\sqrt{5}}$$

In this formula, notice that $|-\varphi^{-1}| < 1$, and so for large n, this term is very small. Thus we have the approximation $a_n \approx \varphi^{n+1}/\sqrt{5}$, which gives us a good picture of the rate of growth of the Fibonacci sequence: each term is a about φ times the previous one.

A similar analysis can be applied to other recurrence relations. An equation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + + \dots c_r a_{n-r}$$

can be written as a matrix equation

$$\begin{bmatrix} a_{n+1} \\ \vdots \\ a_{n+r} \end{bmatrix} = \begin{bmatrix} 0 & 1 & & & \\ & 0 & 1 & & \\ & & \ddots & \ddots & \\ & & & 0 & 1 \\ c_r & & \dots & & c_1 \end{bmatrix} \begin{bmatrix} a_n \\ \vdots \\ a_{n+r-1} \end{bmatrix}$$

If we can diagonalize the matrix appearing in this equation, we can find an explicit formula for a_n.

3 Linear Dynamical Systems in Nature

A method similar to our analysis of a recurrence relation can be applied to many systems which evolve in discrete steps according to some linear function. As an example, consider a species of beetle with the following life history. A juvenile beetle, in its first year of life, produces no offspring, and half of all beetles die in this period. In its second year of life, each beetle produces on average 2 offspring, and again half die. In its third year, a beetle produces an average of 1 offspring and dies of old age.

Question. Do we expect the population of beetles to grow or decline? At what rate?

Question. What distribution of ages do we expect to see in the beetle population?

To answer these questions, we can describe the state of the population and its evolution as follows. In a given year n, the population has x_n juveniles, y_n beetles in their second year, and z_n beetles in their third year. We can package these numbers into a vector

$$v_n = \begin{bmatrix} x_n \\ y_n \\ z_n \end{bmatrix}$$

called the *state vector*. The population changes from year to year according to the equation

$$v_{n+1} = \begin{bmatrix} x_{n+1} \\ y_{n+1} \\ z_{n+1} \end{bmatrix} = \begin{bmatrix} 0 & 2 & 1 \\ 1/2 & 0 & 0 \\ 0 & 1/2 & 0 \end{bmatrix} \begin{bmatrix} x_n \\ y_n \\ z_n \end{bmatrix} = Av_n$$

The matrix appearing here is called the *transition matrix* of this dynamical system. Given a starting state v_0, the population after t years is described by $v_n = A^n v_0$. We can get more clarity as to what to expect by diagonalizing A. The characteristic polynomial is

$$\det \begin{bmatrix} -t & 2 & 1 \\ 1/2 & -t & 0 \\ 0 & 1/2 & -t \end{bmatrix} = 1(1/4 - 0) - t(t^2 - 1) = -t^3 + t + 1/4$$

This has roots $\lambda_1 \approx 1.1$, $\lambda_2 \approx -0.3$ and $\lambda_3 \approx -0.8$. Let v_1, v_2, v_3 be eigenvectors for these eigenvalues. Then given an initial state vector v_0, we can express it as a linear combination

$$v_0 = a_1 v_1 + a_2 v_2 + a_3 v_3$$

of the eigenvectors. We then have the following formula for v_n:

$$v_n = a_1 \lambda_1^n v_1 + a_2 \lambda_2^n v_2 + a_3 \lambda_3^n v_3$$

Because $|\lambda_2|, |\lambda_3| < 1$, these terms become negligible as n increases, and so we can approximate

$$v_n \approx a_1 \lambda_1^n v_1$$

Because $\lambda_1 \approx 1.1$, this tells us that in the long run, the beetle population grows by about 10% per year.

Remark. For this to be true, we need $a_1 > 0$. In our language, a_1 is the first entry in $S^{-1} v_0$. It turns out that if one chooses v_1 to have positive entries, then all entries in the top row of S^{-1} will be positive, so this is true for any positive initial population.

In the long run, the age distribution in the population is expected to look like that of v_1. As a reminder, there are infinitely many vectors v_1 we could use here, all scalar multiples of each-other. To interpret it as an age distribution, we want a v_1 whose entries add up to 1. In our case this is

$$v_1 \approx \begin{bmatrix} 0.6 \\ 0.3 \\ 0.1 \end{bmatrix}$$

so we expect to see about 60% of the population be juvenile, 30% be in their second year, and 10% in their third year.

Remark. A limitation of this description is that in an ecosystem with limited resources, we expect competition to drive down the probability of survival as the population becomes large, leading to a population which levels off rather than growing exponentially forever. To describe this mathematically would require a nonlinear function; our linear description could be a well justified approximation on a time scale where the population is small relative to the carrying capacity of the environment, for instance if the species spreads into a new ecosystem.

In general, a *linear dynamical system* consists of an initial state vector v_0, and a transition matrix A, such that subsequent states follow the rule

$$v_{k+1} = A v_k$$

and so

$$v_k = A^k v_0$$

for all k. Systems like this appear throughout the sciences, usually as approximation descriptions of some system that evolves over time.

By diagonalizing A,

$$A = S \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} S^{-1}$$

we have the formula

$$v_k = S \begin{bmatrix} \lambda_1^k & & \\ & \ddots & \\ & & \lambda_n^k \end{bmatrix} S^{-1} v_0$$

The eigenvalues λ_i tell us a lot about the long term behavior of the system. For example, if $|\lambda_i| < 1$ for all i, then v_k will approach 0 as $k \to \infty$. In general, the λ_i with $|\lambda_i|$ larger than all the others, if one exists, will "dominate" the long term behavior of the system. As $k \to \infty$, the state vector v_k will become approximately proportional to an eigenvector for this dominant eigenvalue (assuming it had a nonzero coefficient to begin with.)